

CBCS SCHEME

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18EC35

Third Semester B.E. Degree Examination, Jan./Feb. 2021
Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. Write neat diagrams wherever necessary.*

Module-1

- 1 a. With a neat diagram, describe the functional units of a computer. Give few examples for I/O devices. (06 Marks)
- b. Discuss IEEE standard for single-precision and double-precision floating point numbers, with standard notations. (06 Marks)
- c. Develop an Assembly Language Program (ALP) for the expression $Y = Ax^2 + Bx + D$ using 3-address, 2-address and 1-address instruction formats. Assume A, B, C, D, Y as memory locations and x as immediate data. (08 Marks)

OR

- 2 a. With a neat diagram, discuss the operational concepts in a computer highlighting the role of PC, MAR, MDR and IR. (08 Marks)
- b. Perform subtraction on the following pairs of numbers using 5-bit signed 2's-complement format. Indicate about overflow in each case:
 i) +10 and -8 ii) +12 and +9 iii) -15 and -9 iv) -14 and +5 (08 Marks)
- c. Distinguish between Big-endian and little-endian memory assignment. With a neat sketch, show how the number 26789435 is stored using these methods. (04 Marks)

Module-2

- 3 a. Define addressing mode. Explain any four basic addressing modes with syntax and examples. (08 Marks)
- b. What is subroutine? With a pseudocode or program segment, illustrate parameter passing using registers. (06 Marks)
- c. Consider a database of marks scored by students in 3 tests, stored in memory starting at address LIST. Each student record consists of studentID followed by marks in 3 tests. Assume each of these to be 4 bytes in size. There are 50 students in the class and this value is stored at location NUM.
 i) Sketch the memory map showing all details
 ii) Develop an ALP using Indexed Addressing mode, to compute the sum of scores by all the students in Test2 and store the result in location SUM. Write appropriate comments. (06 Marks)

OR

- 4 a. Discuss Auto-increment and Auto-decrement addressing modes with syntax. Consider a set of numbers (each 4 bytes in size) stored in memory starting at address TABLE. Total numbers are N and this value is stored at location LOCN.
 i) Sketch the memory map showing all details
 ii) Develop an ALP using Auto-increment addressing mode, to compute the sum of all numbers and store the result at memory address RESULT. Write appropriate comments. (08 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
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- b. Define stack. Explain PUSH and POP operations on stack with neat sketches and examples. (06 Marks)
- c. Consider a register R1 to size 16-bits with initial data 5867_h. With neat sketches, depict the output in each case, after performing the following operations:
 i) LshiftL #2, R1 ii) AshiftR #1, R1 iii) RotateR #1, R1
 Note: For each operation, R1 value is to be taken as 5867_h and carry flag is indicated cleared. (06 Marks)

Module-3

- 5 a. Distinguish between memory mapped I/O and standard I/O. Write a program segment to read a line of text from keyboard and display it. (08 Marks)
- b. What is interrupt priority? Why is it necessary? With relevant diagram, discuss daisy-chain method of handling multiple interrupt requests. (06 Marks)
- c. Explain distributed arbitration mechanism in DMA with a neat diagram. (06 Marks)

OR

- 6 a. With a neat diagram, discuss implementation of interrupt priority using individual request and acknowledge lines. (06 Marks)
- b. Briefly explain: i) Vectored interrupts and ii) Registers in a DMA interface. (06 Marks)
- c. Explain centralized arbitration mechanism in DMA with a neat sketch and timing diagram. (08 Marks)

Module-4

- 7 a. Classify memory in a computer. With a neat diagram, describe the organization of 2M × 8 DRAM chip. (08 Marks)
- b. What is cache memory? Explain direct mapping technique with a neat diagram. (08 Marks)
- c. Briefly discuss the concept of virtual memory with a diagram. (04 Marks)

OR

- 8 a. Briefly explain the working of 1-bit CMOS SRAM cell with a schematic. (06 Marks)
- b. What is mapping function? Explain set-associative cache mapping technique with a relevant diagram. (08 Marks)
- c. With a neat diagram, explain the principle of working of magnetic disk. (06 Marks)

Module-5

- 9 a. Explain single-bus organization of data path in a processor with a neat diagram. Highlight the importance of gating signals. (08 Marks)
- b. Develop the complete control signal sequence for the instruction Add (R1), R3 with appropriate remarks. (06 Marks)
- c. Discuss micro programmed control unit design with relevant diagrams. (06 Marks)

OR

- 10 a. List different ways of improving CPU performance. With a neat diagram, discuss three-bus organization of CPU. Compare the performance with single-bus organization. (08 Marks)
- b. Discuss Hardwired control unit organization with relevant diagrams and illustrate the logic to generate Z_{in} control signal. (08 Marks)
- c. Define the following:
 i) Gating signal ii) Control word iii) Microroutine iv) Control store. (04 Marks)

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Third Semester B.E. Degree Examination, Dec.2019/Jan.2020
Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, explain basic operational concept of computer. (10 Marks)
- b. Explain in brief different types of key parameters that affect the processor performance. (05 Marks)
- c. Explain the Bus Structures. (05 Marks)

OR

- 2 a. Illustrate Instruction and Instruction sequencing with an example. (10 Marks)
- b. Define Byte Addressability, Big-endian and Little-endian assignment. (06 Marks)
- c. Represent 85.125 in IEEE floating point using single precision. (04 Marks)

Module-2

- 3 a. What is an addressing mode? Explain any five types of addressing modes with example. (10 Marks)
- b. Write a program to add 'n' number using indirect addressing mode. (06 Marks)
- c. Explain various assembler directives used in assembly language program. (04 Marks)

OR

- 4 a. Explain stack operation with an example. (10 Marks)
- b. Explain subroutine linkage with an example using linkage register. (06 Marks)
- c. Explain the shift and rotate operations with example. (04 Marks)

Module-3

- 5 a. Showing the possible register configuration in I/O interface, explain program controlled input/output. (10 Marks)
- b. What is an interrupt? With an example illustrate the concept of interrupt. (10 Marks)

OR

- 6 a. Explain in detail, the situations where a number of devices capable of initiating interrupts are connected to processor. How to resolve the problems? (10 Marks)
- b. Explain the registers involved in a DMA interface, to illustrate DMA. (06 Marks)
- c. Explain the concept of Vectored Interrupt. (04 Marks)

Module-4

- 7 a. With figure, explain Internal Organization of 2M×8 dynamic memory chip. (10 Marks)
- b. Illustrate Internal structure of static memories. (10 Marks)

OR

- 8 a. With a neat diagram, explain virtual memory organization. (10 Marks)
- b. Briefly explain any four non-volatile memory concepts. (05 Marks)
- c. Briefly explain secondary storage devices. (05 Marks)

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Module-5

- 9 a. Explain the three-bus organization of the processor and its advantages.
b. Discuss the organization of hardwired control unit.
c. Discuss the control sequence for execution of instruction $ADD(R_3), R_1$

(10 Marks)

(05 Marks)

(05 Marks)

OR

- 10 a. With a block diagram, describe the organization of a micro programmed control unit.
b. Describe the sequence of control signals to be generated to fetch an instruction from memory in a single bus organization.

(10 Marks)

(10 Marks)

(10 Marks)

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Third Semester B.E. Degree Examination, Aug./Sept.2020

Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the operation of computer with neat block diagram. (10 Marks)
- b. Explain computer basic performance equation. (04 Marks)
- c. Explain following with an example : i) Three – address instruction
ii) Two – address instruction iii) One – address instruction. (06 Marks)

OR

- 2 a. Explain Single –BUS structure in computer. (06 Marks)
- b. Explain system software functions in computer. (06 Marks)
- c. What is Operating system? Explain user program and OS routine sharing the processor. (08 Marks)

Module-2

- 3 a. Explain Big-Endian and Little-Endian with neat diagram. (08 Marks)
- b. Explain memory operations with examples. (04 Marks)
- c. Explain condition codes with examples. (08 Marks)

OR

- 4 a. Discuss following addressing modes with example :
i) Immediate ii) Register iii) Direct iv) Indirect v) Index. (10 Marks)
- b. What are assembler directive? Explain any five assembler directives. (10 Marks)

Module-3

- 5 a. With a neat diagram, explain how to interface printer to the processor. (10 Marks)
- b. Define Interrupt. Point out and explain the various ways of enabling and disabling interrupts. (10 Marks)

OR

- 6 a. Explain the following method of handling interrupts from multiple devices.
i) Daisy chain method ii) Priority structure. (10 Marks)
- b. Explain operation of DMA with neat diagram. (10 Marks)

Module-4

- 7 a. Explain internal organization of 16×8 memory chip. (10 Marks)
- b. Discuss a single-transistor dynamic memory cell. (06 Marks)
- c. Write a note on Virtual Memory. (04 Marks)

OR

- 8 a. Draw and explain the internal organization of $2M \times 8$ asynchronous DRAM Chip. (08 Marks)
- b. Describe the principles of magnetic disk. (06 Marks)
- c. What is mapping? Explain set associative cache mapping techniques. (06 Marks)

Module-5

- 9 a. Discuss with neat diagram, the single bus organization of data path inside a processor. (10 Marks)
- b. What are the action required to execute a complete instruction $ADD(R3), R1$. (10 Marks)

OR

- 10 a. Draw and explain multiple bus organization of CPU. (10 Marks)
- b. Draw and explain organization of the control unit to allow conditional branching in the microprogram. (10 Marks)

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18EC35

Third Semester B.E. Degree Examination, Feb./Mar. 2022
Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the basic operational concept between the processor and memory with neat block diagram. (08 Marks)
- b. Explain the various parameters affecting the performance of a computer and also provide the basic performance equation. (08 Marks)
- c. Write a short note on single bus structure with neat diagram. (04 Marks)

OR

- 2 a. List out and explain the three systems used for representing signed numbers and also brief about the modular number system concept. (08 Marks)
- b. Explain IEEE standard used for single and double precision floating point number representation with examples. (08 Marks)
- c. Write a short note on Big-endian and little-endian assignment. (04 Marks)

Module-2

- 3 a. What is addressing mode? Explain any four addressing modes with examples. (08 Marks)
- b. What are assembler directives? Explain about the various directives used in the program with example. (08 Marks)
- c. Write a short note on the assembly and execution of programs. (04 Marks)

OR

- 4 a. With neat diagram and program example, explain a simple I/O task between processor, keyboard and display. (10 Marks)
- b. What is subroutine? Illustrate the subroutine function with parameter passing by value and reference with suitable program. (10 Marks)

Module-3

- 5 a. Explain the concept of memory mapped I/O with neat diagram of I/O interface with program example. (10 Marks)
- b. Write short notes on: (i) Interrupt hardware (ii) Interrupt nesting (10 Marks)

OR

- 6 a. What is an interrupt? Explain about various implementation techniques of interrupt. (10 Marks)
- b. Explain how simultaneous interrupt request is handled using the concept of Daisy Chain. (10 Marks)

Module-4

- 7 a. Explain the internal organization of memory chips with example. (08 Marks)
- b. Explain the internal organization of $2M \times 8$ DRAM chip with neat diagram. (08 Marks)
- c. Write a short note on ROM. (04 Marks)

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OR

- 8 a. Discuss about the use of cache memory in the processor system.
b. What is virtual memory? Explain its organization with neat diagram.
c. Write a short note on magnetic hard disk.

(08 Marks)
(08 Marks)
(04 Marks)

Module-5

- 9 a. Explain single-bus organization of the data path inside a processor with neat diagram.
b. Explain the process of fetching a data word from memory using respective registers of a processor with neat diagram.

(10 Marks)
(10 Marks)

OR

- 10 a. Explain the control signal generation required for proper sequence of instructions in the processor.
b. What is microprogrammed control? Explain its basic organization with suitable diagram and example.

(10 Marks)
(10 Marks)

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18EC35

Third Semester B.E. Degree Examination, June/July 2023
Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with a neat diagram, the basic Operational concept of a Computer. (08 Marks)
 b. Explain how to measure the performance of a Computer. (06 Marks)
 c. Write a note on Types of Computers. (06 Marks)

OR

- 2 a. Explain IEEE standard for Floating point number. (08 Marks)
 b. Explain the methods to improve the performance of Computer. (08 Marks)
 c. Write a note on Processor clock. (04 Marks)

Module-2

- 3 a. What is an Addressing Mode? Explain any four addressing mode with an example. (10 Marks)
 b. With an example, explain the concept of BIG – ENDIAN and LITTLE – ENDIAN Assignment of Memory Storage. (10 Marks)

OR

- 4 a. Explain the concept of Stacks and Queues. (08 Marks)
 b. What are Assembler directives? Explain the various assembler directives with examples. (08 Marks)
 c. With an example, explain Shift and Rotate Instructions. (04 Marks)

Module-3

- 5 a. Define Interrupt. Explain Daisy chain and Priority Structure methods of handling interrupts from multiple devices. (10 Marks)
 b. With a neat diagram, explain DMA Controller Operation with its Interface Registers. (10 Marks)

OR

- 6 a. Define Exceptions. Explain the different types of Exceptions. (06 Marks)
 b. Explain the Tree structure of USB with Split bus operation. (06 Marks)
 c. With a neat diagram, explain Centralized and distributed bus arbitration schemes. (08 Marks)

Module-4

- 7 a. Define Cache Memory. Explain various types with neat diagram. (08 Marks)
 b. Write a note on Classification of a Memory Structure. (04 Marks)
 c. Define the following terms :
 i) Memory Latency ii) Memory Bandwidth
 iii) Memory Access time iv) Memory Cycle time. (08 Marks)

OR

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- 8 a. Explain with block diagram, the Operation of SD RAM. (10 Marks)
b. Define ROM Point out and explain various types of ROM's. (10 Marks)

Module-5

- 9 a. Explain with neat diagram, Single Bus Organisation of data path inside a processor. (10 Marks)
b. What are the actions required to execute a Complete Instruction Add(R3), R1? (10 Marks)

OR

- 10 a. Explain Hardwired Control Unit Organisation. (10 Marks)
b. Explain Multiple bus / three bus Organization, with a neat diagram. (10 Marks)